

A Resource-Efficient Content Sharing Mechanism in Large-Scale UAV Named Data Networking

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Abstract—In recent years, there has been significant attention in UAV Named Data Networking (UNDN) from both industry and academia. This network paradigm adopts a “request-reply” communication model that allows UAVs to access desired content without the need for specific information regarding the geographical location or IP address of the content producer. This IP-independent design is well-suited for dynamic UAV swarms, but it presents challenges in establishing matching policies between content consumers and producers. This is because that during the distributed decision-making process in content sharing, consumers cannot possess private information regarding producers, and producers may lack the motivation to distribute content. As a result, a revelation and incentive mechanism is needed to be formulated in the system. In this paper, a resource-efficient content-sharing mechanism is proposed to address the aforementioned challenges. First, we propose a contract-based mechanism to incentivize content producers to share content and reveal their private information at the same time. The problem of obtaining the optimal contract is discussed in both cases of information asymmetry and complete information. Then, the Gale-Shapley (GS) algorithm is adopted to make a stable many-to-one matching between content consumers and content producers. The simulation results verify the feasibility, effectiveness and energy efficiency of the proposed mechanism.

Index Terms—UAV named data networking (UNDN), content sharing, contract theory, matching theory.

I. INTRODUCTION

RECENTLY, the widespread application of UAV swarms has brought tremendous benefits and changes to our lives, industry, and military affairs [1]. The members of the

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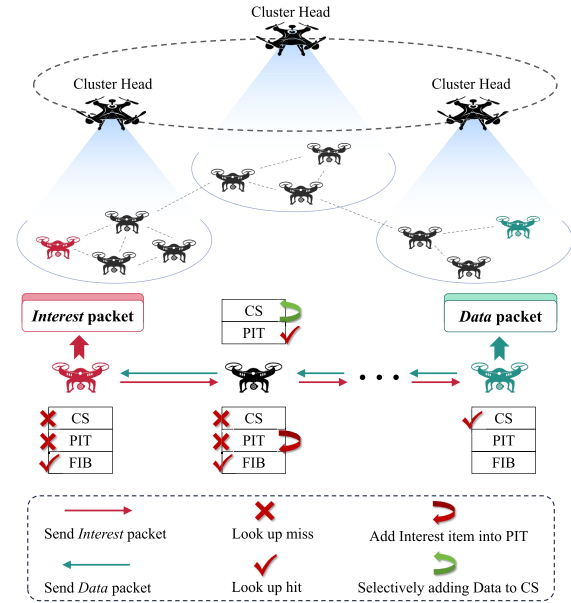


Fig. 1. Communication paradigm in UAV named data networking (UNDN).

UAV swarm utilize a new architecture for collaboration to complete complex tasks, known as the flying ad-hoc network (FANET) [2]. In FANET, all UAV nodes form an ad hoc network, where each node can directly communicate with other nodes by establishing end-to-end connections based on unique IP addresses. However, the high mobility property of UAVs always leads to frequent network topology changes [3], making it difficult to maintain stable end-to-end connectivity in such a highly dynamic network. Additionally, with the increase in the number of UAV nodes in the network, assigning IP addresses to each node is also challenging, especially considering the issue of IP address depletion and the high mobility characteristics of UAVs [4]. Therefore, IP-based protocols are not suitable enough for FANET due to the above issues.

In recent years, a new architecture known as Named Data Networking (NDN) has offered an effective solution to the aforementioned problems [5]. NDN utilizes a “request-reply” communication model by sending two types of packets. Specifically, when a UAV node intends to request data, it broadcasts an *Interest* packet with content name as *content consumer*. Upon receiving the *Interest* packet, a UAV node checks if it has the requested content. If the content is present, the node serves as *content producer* and returns the *Data*

packet back to the content consumer. This network architecture enables UAVs to access desired content without requiring knowledge of the geographic location or IP address of the content producer. This is particularly well-suited for UAV swarms, as UAVs prioritize the required content itself over the identity of the content producer. The communications architecture of UAV named data networking (UNDN) is shown in Fig. 1.

Despite the above advantages of UNDN, it still has some issues that need to be studied, such as forwarding strategy, caching mechanism, naming scheme, etc. In forwarding strategy, the problem of *Interest* flooding has been widely studied. However, issues in the content returning process are rarely involved. Specifically, in UNDN, after a content consumer broadcasts the *Interest* packet, all content producers will respond by sending the *Data* packet back to content consumer, which can cause serious resource waste, data redundancy, and even network congestion. This situation is more severe in resource-constrained UAV swarm networks. Therefore, it is essential to carry out a more precise matching mechanism between these two sides for the sake of resource efficiency. Nevertheless, how to make distributed decisions and achieve accurate matching in UNDN is the core challenge. In particular, content consumers lack access to private information (e.g., reputation value, residual battery) of content producers, which makes it difficult to judge the quality of nodes to make decisions. In addition, content producers lack motivation to share content due to selfishness. Therefore, it is essential to design an efficient incentive mechanism.

In this paper, we are motivated to establish a resource-efficient two-stage content-sharing mechanism using contract theory [6] and matching theory [7]. In the first phase, to incentivize content producers to send content while also exposing their private information, each content consumer designs a set of contracts acting as principal for content producers, who function as agents. Each contract item contains bundled content amounts and corresponding rewards. Then, content consumers broadcast the contract items with *Interest* packets and each content producer selects the item that can maximize its utility. The application of contract theory offers two distinct advantages in such scenarios. Firstly, it can motivate content producers in situations of asymmetric information. Secondly, it can reveal the types of content producers (i.e., private information), ensuring that content producers must be honest and cannot misrepresent their types. In the second stage, a precise distributed many-to-one matching game is conducted between two sides, which implies that one content consumer can only match with one content producer, while one content producer can provide content to multiple content consumers. During the matching process, both content producers and content consumers first recalculate their utilities based on the results of the selected contracts from the first stage to formulate preference lists. Subsequently, the matching process commences in a distributed manner.

The major contributions of the proposed work are summarized as follows:

- We introduce the NDN network architecture into UAV swarm networks, utilizing a “request-reply” communica-

tion model and a content-centric routing mechanism to address the issue of frequent interruptions of end-to-end connections in IP-based solutions.

- We focus on the issues of redundant data and resource waste during the data packet return process in UNDN. To solve this problem, we propose a resource-efficient content sharing mechanism. Specifically, we adopt the Gale-Shapley (GS) algorithm [8] for distributed many-to-one precise matching between content producers and content consumers to address the above issues. To the best of our knowledge, we are the first to study this issue in UNDN.
- We design a contract-based revelation and incentive mechanism to enable both parties to establish preference lists during the matching process while encouraging content producers to provide content. Both information asymmetry and complete information situations are considered in this paper.
- We validate our designed contract items and the distributed many-to-one matching method through simulations experiments. Our proposed mechanisms exhibit superiority in terms of content contribution, social welfare and so on. Additionally, we compared our designed content transmission mechanism with the original NDN content transmission mechanism, and the experimental results demonstrate that our proposed mechanism effectively reduces the energy consumption of content transmission.

The rest of this paper is structured as follows. A comprehensive review of NDN, contract theory, and matching theory is provided in Section II. The system model of UNDN is illustrated in Section III. In Section IV, a contract-based incentive mechanism is designed and the optimal contract items are derived. Then, the matching process is discussed in Section V. The simulation results and analysis are presented in Section VI. Finally, we summarize this article in Section VII.

II. RELATED WORKS

A. Named Data Networking

The emergence of Content-Centric Networking (CCN) [9] has propelled the shift in network interaction from traditional “data transmission” to “meaningful communication”. In this context, the Named Data Networking (NDN) architecture is proposed. Recently, NDN has been widely used in ad hoc networks to solve the problems of IP-based solutions.

A comprehensive survey is provided in [10], covering an overview of NDN and VANET, realizations, architectures, naming schemes, forwarding strategies, caching schemes, mobility support, security solutions, simulation tools, and future research directions. In [11], Ahmed et al. propose a scheme called “CODIE” to control broadcast storms in vehicular NDN (VNDN). Similarly, Grassi et al. propose a location-based forwarding mechanism named “Navigo” to alleviate *Interest* packets flooding in VNDN [12]. Besides, Chen et al. suggest an intelligent caching strategy to reduce the delay of data acquisition [13]. Hou et al. in [14] propose

TABLE I
SUMMARY OF KEY NOTATIONS

Notation	Definition	Notation	Definition
$\mathcal{M}$	Set of content producers: $\mathcal{M} = \{1, \dots, j, \dots, M\}$	$\mathcal{N}$	Set of content consumers: $\mathcal{N} = \{1, \dots, i, \dots, N\}$
R_i	Reputation value of content producer i	B_i	Residual battery of content producer i
B_i^{res}	Current battery capacity of content producer i	B_{max}	Maximum battery capacity of content producers
q_{ij}	Content Amounts provided by content producer i to content consumer j	q_j^{min}	Minimum content amounts that content consumer j can accept
q^{max}	Complete content volume of a specific content	E_{ij}^c	Caching cost for content producer i to cache q_{ij} content
c	Unit caching cost	$d(i, r)$	Distance between content consumer i and next-hop r
E_{ij}^T	Energy Consumption of content producer i for transmitting q_{ij} content	E_{elec}	Unit energy consumption
τ	Distance switching threshold of two models	η_{fs}	Energy consumption factor in free space model
η_{mp}	Energy consumption factor in multi-path fading model	θ_i	Type of content producer i
p_i	Probability of content producer belonging to type θ_i	π_{ij}	Rewards of content producer i for sharing q_{ij} content
$U^P(\theta_i, q_{ij})$	Utility of content producer in contract design i	$U^C(\theta_i, q_{ij})$	Utility of content consumer in contract design j
a	Constant coefficient for gain function $V(q)$	$\mathcal{P}^C$	Preference lists for all content consumers
$\mathcal{P}^P$	Preference lists for all content producers	Φ	Matching function
K_i	Matching quota of content producer i	K_{max}	Maximum number of matches
$U_m^P(\theta_i, q_{ij})$	Utility of content producer i in matching stage	$U_m^C(\theta_i, q_{ij})$	Utility of content consumer j in matching stage
δ_i	Reward for content producer i in matching process	m	Reward coefficient in matching process
λ	Unit transmission energy cost	ϵ	Weighting factor

a method for predicting data packet backhaul (DBPM) using cluster routing in VNDN to improve the communication QoS.

Furthermore, there are also several articles exploring the integration of NDN into UAV swarm networks. Araújo et al. in [15] deploy NDN in FANET to solve the intermittent connectivity problem. The authors further propose a forwarding strategy based on multiple criteria to mitigate *Interest* flooding. Similarly, Qiu et al. [16] integrate NDN to UAV swarm networks and present an integrated host and content-centric routing (IHCR) mechanism, aiming to utilize the strengths of both routing paradigms. However, most of the existing articles focus on the issues during the transmission of *Interest* packets and overlook the issues that exist during the *Data* packet return process, which is the focus of our paper.

B. Contract Theory

Contract theory is considered an effective means to motivate participants under information asymmetry. Currently, many articles apply contract theory in the data market. For instance, in [17], Tian et al. address the problem of data trading in the online data market and introduce a contract-based mechanism. This mechanism aims to maximize the utility of the data seller, who is considered the contract principal in this context. Zhang et al. [18] propose a multi-dimension incentive mechanism based on contract theory to motivate users' participation in the crowdsourcing scenario. Similarly, Ma et al. [19] present a contract-based incentive scheme in Wi-Fi network crowdsourcing. Considering privacy in the data market, Xu et al. [20] assume that buyers can evaluate the privacy of sellers, and contract theory is used to strike a balance between safeguarding privacy and maximizing data utility.

Additionally, contract theory finds extensive application in many other scenarios, such as multi-access edge computing

(MEC), federated learning (FL), and so on. In [21], Su et al. establish a two-step framework among MEC operator, edge computation nodes (ECNs) and computation service subscribers (CSSs), where contract theory is adopted in the first step to stimulate ECNs to offer CPU resources. Zhou et al. [22] propose an incentive mechanism aimed at motivating vehicle service nodes to share resources in vehicular fog computing (VFC). The study in [23] combines VFC with smart EV charging for joint optimization, employing contract theory to provide an optimal charging protocol. Furthermore, in [24], the authors propose a contract-based approach to motivate mobile devices to join FL tasks. Lim et al. [25] also adopt contract theory in the FL system. The model owner formulates a series of multi-dimensional contracts for UAVs, which act as service providers, to find the most suitable UAV for a specific sub-region.

C. Matching Theory

Matching theory can be generally categorized into three groups: one-to-one matching, many-to-one matching, and many-to-many matching [26]. Matching theory can handle the preferences of heterogeneous nodes, which makes it suitable to solve assignment problems. For example, in [25], Lim et al. carry out the many-to-one matching process to assign the optimal UAVs to each sensing sub-region. Similarly, Zhou et al. [27] also use matching theory between UAVs and sub-regions in mobile crowd sensing. Matching theory finds extensive application in resource allocation as well. In [28], Zhou et al. address the energy-efficiency resource allocation problem by employing one-to-one matching between cellular UEs and D2D pairs. Zhang et al. [29] study the resource allocation problem in heterogeneous cloud radio access networks, utilizing many-to-one matching and coalition game.

In MEC scenarios, Du et al. [30] establish a dual-level matching game to optimize computing resource management in small-cell networks. The study in [31] suggests a joint optimization framework involving fog nodes, data service operators, and subscribers, where stackelberg game and many-to-many matching are utilized to achieve optimal resource allocation strategies.

III. SYSTEM MODEL

In this paper, we explore the application of NDN in UAV swarm networks, where UAVs are organized with a cluster structure for effective management. We consider dividing continuous time into multiple discrete time slots and regard the entire system as quasi-static during each time slot. And in each time slot, we assume the presence of M content consumers and N content producers for a specific type of content. The set of content consumers and content producers can be denoted as $\mathcal{M} = \{1, \dots, j, \dots, M\}$ and $\mathcal{N} = \{1, \dots, i, \dots, N\}$, respectively. As shown in Fig. 2, two stages are contained in our system model. In the first stage, each content consumer (principal) designs multiple contract items, which are disseminated along with *Interest* packets. Upon receiving the *Interest* packet, each UAV node first checks if it possesses the corresponding content. A successful content name match signifies that the node contains the relevant content and can serve as a content producer. Each potential content producer (agent) then preselects an item from the contract sets and sends their preselection results back to the cluster head. The cluster head, in turn, transmits this information to neighboring cluster heads, ultimately conveying it back to the content consumer. After the contract preselection phase is completed, the matching between content consumers and content producers begins. In this phase, each content consumer requires only one content producer to provide the content, while each content producer can serve multiple content consumers. Therefore, this phase can be modeled as a many-to-one matching game. The two phases in our system model are stated below:

1) Content Consumers Design Contract

In UNDN, content producers lack motivation to share content with other UAVs without additional incentive mechanisms due to their selfishness. The types of content producers, such as reputation value and remaining electricity, constitute private information that content consumers are unaware of. Therefore, each content consumer designs a set of contract items to stimulate content producers while revealing their corresponding types, thus facilitating the matching process in the second stage.

2) Many-to-One Matching Between Content Consumers and Content Producers

For a content consumer, multiple content producers may have accepted the contracts. If all these nodes send data packets to the content consumer, it can result in content redundancy and lead to a waste of transmission resources. Besides, each content consumer tends to choose a high-quality producer. Therefore, after completing the first phase of contract preselection, both sides establish their preference lists based on their utilities. Then, the GS algorithm is employed to achieve

many-to-one stable matching between content consumers and content producers.

We consider the model of reputation value, residual battery, caching cost, and transmission cost of the UAV as follows.

A. Reputation Value, Residual Battery, Caching Cost Model

In UNDN, when participating in forwarding, UAVs may engage in malicious behaviors, such as discarding packets to save energy, which will result in a deterioration in network performance. Therefore, we propose reputation value R to measure the performance of nodes in sharing content, where $R \in (0, 1]$. The initial reputation value is set to 0.5. For any nodes in the network, forwarding behavior can affect the reputation value. Successful forwarding results in an increase in reputation value, while unsuccessful forwarding leads to a decrease in reputation. The reputation value of the i th UAV is calculated as follows:

$$R_i = \lambda r + (1 - \lambda) R_i^{old}, \quad (1)$$

where r represents the most recent forwarding result, which can have only two possible values: 1 for successful forwarding and 0 otherwise. R_i^{old} denotes the old reputation value of node i . λ is a weight factor between the current forwarding result and the old reputation value.

Residual battery B is another parameter to evaluate the quality of UAVs. A UAV with high residual power is more inclined to forward and share content with other nodes, whereas a UAV with low residual power tends to conserve energy for its own survival. In this paper, the residual battery B of i th UAV is calculated by the ratio of the current battery capacity to the maximum battery capacity. Thus, $B \in (0, 1]$. The definition of B is given as follows:

$$B_i = \frac{B_i^{res}}{B_{\max}}. \quad (2)$$

The caching cost of UAV is incurred due to cache contents, which depends on the size of cached contents. The caching cost of content producer i when caching q_{ij} contents for content consumer j is defined as follows:

$$E_{ij}^c = c \times q_{ij}, \quad (3)$$

where c is the unit caching cost.

B. Energy Consumption Model

This paper considers the energy consumption of UAVs due to content transmission. The transmission energy consumption E^T stems from the transmitter's circuitry and power amplifier. Regarding the issue of wireless channel fading, we are taking into account both the free space model and the multi-path fading model simultaneously. The energy consumed by content producer i when transmitting q_{ij} content to content consumer j can be formulated as [32]:

$$E_{ij}^T = \begin{cases} q_{ij} \times [E_{elec} + \eta_{fs} \times d(i, r)^2], & d(i, r) < \tau \\ q_{ij} \times [E_{elec} + \eta_{mp} \times d(i, r)^4], & d(i, r) \geq \tau \end{cases}, \quad (4)$$

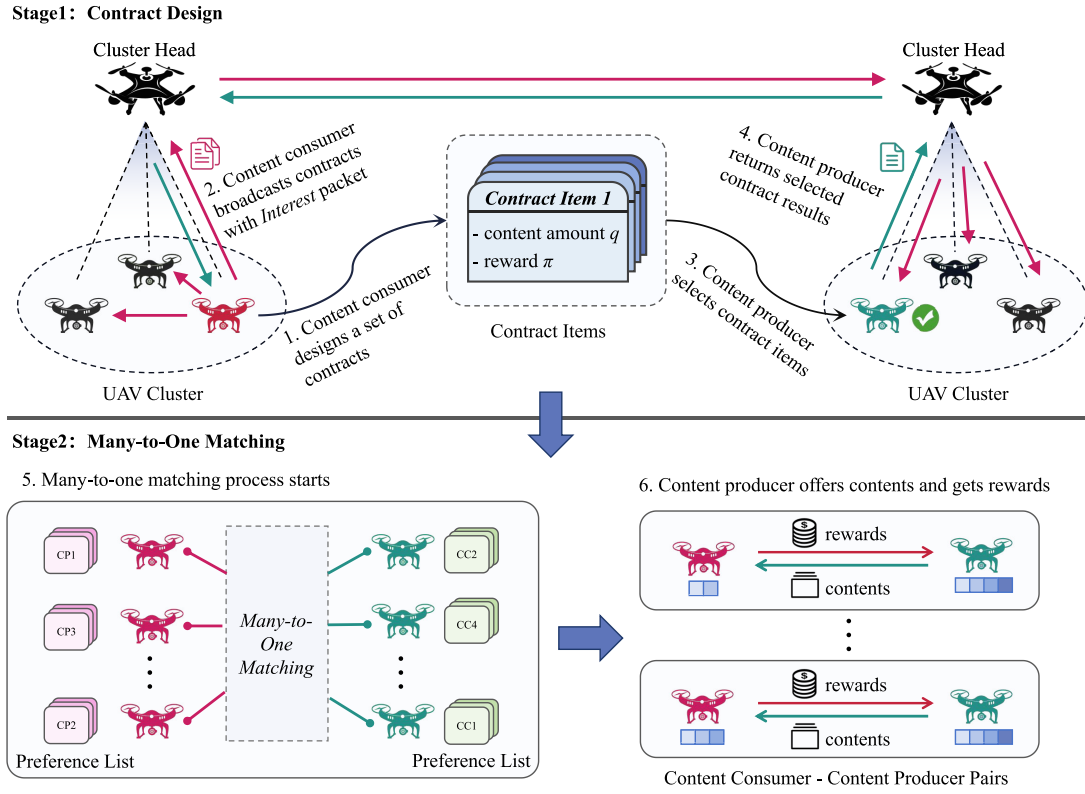


Fig. 2. System model.

where r is the next-hop node of content producer i . Due to the high mobility nature of UAVs, transmitting data packets to content consumers along the *Interest* packet's reverse paths is no longer applicable in UNDN. Consequently, in this paper, the content producer initially transmits the *Data* packet to the cluster head, which then sequentially relays it to the adjacent cluster heads, ultimately delivering it to the content consumer. E_{elec} represents the unit energy consumption. η_{fs} and η_{mp} represent the energy consumption factors for power amplifiers in the free space model and multi-path fading model, respectively. $d(i, r)$ is the distance between content producer i and the next-hop node r , and τ denotes the distance switching threshold of the two aforementioned models:

$$\tau = \sqrt{\frac{\eta_{fs}}{\eta_{mp}}}. \quad (5)$$

IV. CONTRACT-BASED INCENTIVE MECHANISM DESIGN

In this part, a contract-based disclosure and incentive mechanism is designed to expose private information of content producers to content consumers while motivating content producers to share content. We first present the type model of the content producer. Then, we model the utility functions of both content consumers and content producers, and formulate the optimization problem. Lastly, we derive the optimal contracts both with and without information asymmetry.

A. Content Producer Type Modeling

During the content sharing process in UNDN, sending content will cause power consumption, and the trustworthiness

of the node also requires attention. As a result, type θ is introduced to denote the willingness and ability of nodes to share content as content producers. The two parameters mentioned above, reputation value R and residual battery B , are the confidential information of the node, which is only known to itself. Inspired by [33], the type of content producer is determined by both reputation value R and residual battery B :

$$\theta_i = \alpha_1 R_i + \alpha_2 B_i, \quad (6)$$

where $\alpha_1, \alpha_2 \in (0, 1)$ are weight factors that regulate the importance of R and B , and $\alpha_1 + \alpha_2 = 1$. A higher θ indicates that the corresponding content producer has a greater reputation value and remaining battery capacity, thus is more willing to share content and also deserves more rewards. We arrange the types of content producers in ascending order and assume their types are discretely finite, which is represented by

$$\theta_1 < \dots < \theta_i < \dots < \theta_N. \quad (7)$$

In this paper, the expression “content producer i ” is equivalent to “type θ_i content producer”. Moreover, the probability of a content producer belonging to type θ_i is expressed as p_i , and $\sum_{i=1}^N p_i = 1$.

B. Contract Formulation

In the contract-based model, for a specific content, each content consumer designs a set of contribution-reward bundle contracts (q, π) for different types of content producers. The contract collection designed by content consumer j is (q_j, π_j) . Each contract item is (q_{ij}, π_{ij}) , where q_{ij} represents

the amount of content provided by content producer i to content consumer j , π_{ij} represents the corresponding reward. These contracts are spread with *Interest* packets. For content producers, they can choose to accept or refuse the contracts.

Next, we define the utility functions of content producers and content consumers, each comprising two parts: revenue and cost.

Definition 1 (Utility of Content Producer): The utility of θ_i -type content producer when provides q_{ij} content to content consumer j can be expressed as reward minus caching cost:

$$U^P(\theta_i, q_{ij}) = \theta_i W(\pi_{ij}) - E_{ij}^c. \quad (8)$$

In (8), $W(\pi_{ij})$ is a reward valuation function for the content producer, and it must follow: $W(0) = 0$, $\frac{dW}{d\pi} > 0$, $\frac{d^2W}{d\pi^2} < 0$, which means the reward valuation function is strictly increasing and concave. Without loss of generality and inspired by [4], [21], and [34], we set $W(\pi) = \ln(1 + \pi)$ for the characteristic of decreasing marginal utility.

Definition 2 (Utility of Content Consumer): The utility of content consumer j when receives content from content producer i can be expressed as the value of content minus the reward:

$$U^C(\theta_i, q_{ij}) = V(q_{ij}) - b\pi_{ij}, \quad (9)$$

In (9), $V(q_{ij})$ denotes the gain from content q_{ij} , and there exists $V(0) = 0$, $\frac{dV}{dq} > 0$, $\frac{d^2V}{dq^2} < 0$, which means content gain function is strictly increasing and concave. Furthermore, inspired by [20], $V(q)$ is defined as $V(q) = \sqrt{aq}$ for the characteristic of decreasing marginal utility, in which the positive factor a measures the value of the content to content consumers. b represents the unit cost of content consumers on offering rewards. For simplicity's sake, we assume $b = 1$.

C. Contract Conditions Analysis and Problem Formulation

In this paper, the type of content producer θ is private information, and some nodes may misreport their types to obtain higher rewards. To truly reveal the hidden information (types) of selfish content producers, the feasible contract needs to meet two constraints: Individual Rationality (IR) and Incentive Compatibility (IC).

Definition 3 (IR Constraints): The IR constraints indicate that the contract chosen by a content producer must ensure its utility is non-negative, i.e.,

$$U^P(\theta_i, q_{ij}) = \theta_i \ln(1 + \pi_{ij}) - cq_{ij} \geq 0. \quad (10)$$

Definition 4 (IC Constraints): The objective of each content producer is to maximize its utility. The IC constraints indicate that for a specific θ_i type content producer, its utility can be maximized only when it selects the contract corresponding to its type, i.e.,

$$U^P(\theta_i, q_{ij}) \geq U^P(\theta_i, q_{i'j}), \forall j, \forall i' \neq i, \quad (11)$$

$$\theta_i \ln(1 + \pi_{ij}) - cq_{ij} \geq \theta_i \ln(1 + \pi_{i'j}) - cq_{i'j}, \forall j, \forall i' \neq i. \quad (12)$$

Therefore, the contract-based incentive mechanism inherently reveals the true information of the participants, where content consumers can infer the type of content producer by observing the contract it chooses to accept.

In contract theory, the goal is to maximize the contract principal's expected utility while meeting the constraints of IR and IC. Thus, the problem is formulated as follows:

$$\mathbf{P1}: \max_{q_{ij}, \pi_{ij}} \sum_{i=1}^N p_i (\sqrt{aq_{ij}} - \pi_{ij}), \quad (13)$$

$$\text{s.t. } q_j^{\min} \leq q_{ij} \leq q^{\max}, \quad (13a)$$

$$IR: \theta_i \ln(1 + \pi_{ij}) - cq_{ij} \geq 0, \quad (13b)$$

$$IC: \theta_i \ln(1 + \pi_{ij}) - cq_{ij} \geq \theta_i \ln(1 + \pi_{i'j}) - cq_{i'j}. \quad (13c)$$

(13a) indicates that for a specific type of content, content consumer j can accept a minimum amount of content $q_j^{\min}$, and the total data volume of that content is $q^{\max}$. (13b) is IR constraints, which ensures that each type of content producer can benefit from data sharing. (13c) is IC constraints, which means that each content producer can only maximize its utility by selecting the contract item that corresponds to its type. (13b) and (13c) jointly guarantee the feasibility of contracts.

D. Properties of Feasible Contract

The original contract-based optimization problem **P1** presented in IV-C contains N IR constraints and $N(N-1)$ IC constraints. Because of the complexity of the constraints, the problem becomes too complicated to solve directly. Hence, we employ feasibility conditions to simplify the constraint conditions.

Lemma 1: $\forall j \in \mathcal{M}, \forall i, i' \in \mathcal{N}$ and $i \neq i'$, any feasible contract items satisfy the following conditions [35]:

- 1) If $\theta_i > \theta_{i'}$, then $\pi_{ij} \geq \pi_{i'j}$;
- 2) If $\pi_{ij} > \pi_{i'j}$, then $\theta_i \geq \theta_{i'}$.

Proof:

- 1) $\theta_i > \theta_{i'} \rightarrow \pi_{ij} \geq \pi_{i'j}$.

According to the IC constraints, we have:

$$\theta_i \ln(1 + \pi_{ij}) - cq_{ij} \geq \theta_i \ln(1 + \pi_{i'j}) - cq_{i'j}, \forall i' \neq i, \quad (14)$$

$$\theta_{i'} \ln(1 + \pi_{i'j}) - cq_{i'j} \geq \theta_{i'} \ln(1 + \pi_{ij}) - cq_{ij}, \forall i' \neq i. \quad (15)$$

Then, add (14) and (15), we have:

$$\begin{aligned} & \theta_i \ln(1 + \pi_{ij}) + \theta_{i'} \ln(1 + \pi_{i'j}) \\ & \geq \theta_i \ln(1 + \pi_{i'j}) + \theta_{i'} \ln(1 + \pi_{ij}), \forall i' \neq i. \end{aligned} \quad (16)$$

After transforming, we have:

$$\ln(1 + \pi_{ij})(\theta_i - \theta_{i'}) \geq \ln(1 + \pi_{i'j})(\theta_i - \theta_{i'}), \forall i' \neq i. \quad (17)$$

As $\theta_i > \theta_{i'}$, we can easily derive:

$$\theta_i - \theta_{i'} > 0. \quad (18)$$

By simultaneously eliminating $(\theta_i - \theta_{i'})$ on both sides, we can obtain $\ln(1 + \pi_{ij}) \geq \ln(1 + \pi_{i'j})$. Since $\ln(\cdot)$ is a strictly increasing function, $\pi_{ij} \geq \pi_{i'j}$ holds.

2) $\pi_{ij} > \pi_{i'j} \rightarrow \theta_i \geq \theta_{i'}$.

By representing (16) in another form, we can obtain:

$$\begin{aligned} & \theta_i[\ln(1 + \pi_{ij}) - \ln(1 + \pi_{i'j})] \\ & \geq \theta_{i'}[\ln(1 + \pi_{ij}) - \ln(1 + \pi_{i'j})], \forall i' \neq i. \end{aligned} \quad (19)$$

Due to $\pi_{ij} > \pi_{i'j}$ and $\ln(\cdot)$ is strictly increasing, it is easy to obtain

$$\ln(1 + \pi_{ij}) - \ln(1 + \pi_{i'j}) > 0. \quad (20)$$

Therefore, we can get $\theta_i \geq \theta_{i'}$.

The proof is finished. $\square$

Lemma 1 indicates that π_{ij} increases with θ_i , which means that a higher type content producer can receive no less rewards than a lower type content producer. We summarize this property to **Proposition 1**.

Proposition 1: (Monotonicity of rewards): For any feasible contract, the reward satisfies:

$$0 \leq \pi_{1j} \leq \dots < \pi_{ij} \leq \dots \leq \pi_{Nj}. \quad (21)$$

Similar to **Lemma 1** and **Proposition 1**, we can obtain **Lemma 2** and **Proposition 2** as follows.

Lemma 2: $\forall j \in \mathcal{M}, \forall i, i' \in \mathcal{N}$ and $i \neq i'$, any feasible contract items satisfy $\pi_{ij} > \pi_{i'j}$ if and only if $q_{ij} > q_{i'j}$, and $\pi_{ij} = \pi_{i'j}$ if and only if $q_{ij} = q_{i'j}$.

Proof:

1) *Sufficiency:* $q_{ij} > q_{i'j} \rightarrow \pi_{ij} > \pi_{i'j}$.

Utilizing the inequality in (14), we can obtain:

$$\theta_i[\ln(1 + \pi_{ij}) - \ln(1 + \pi_{i'j})] \geq c(q_{ij} - q_{i'j}) > 0, \forall i' \neq i. \quad (22)$$

Since $c > 0$, $q_{ij} > q_{i'j}$ and $\ln(\cdot)$ is strictly increasing, then $\pi_{ij} > \pi_{i'j}$.

2) *Necessity:* $\pi_{ij} > \pi_{i'j} \rightarrow q_{ij} > q_{i'j}$.

Then, we can transform the inequality (15) into the following form:

$$\theta_i[\ln(1 + \pi_{i'j}) - \ln(1 + \pi_{ij})] \geq c(q_{i'j} - q_{ij}), \forall i' \neq i. \quad (23)$$

Since $\pi_{i'j} < \pi_{ij}$ and $\ln(\cdot)$ is strictly increasing, we can acquire:

$$0 > \theta_{i'}[\ln(1 + \pi_{i'j}) - \ln(1 + \pi_{ij})] \geq c(q_{i'j} - q_{ij}), \forall i' \neq i. \quad (24)$$

And there exists $c > 0$, therefore we have:

$$q_{i'j} - q_{ij} < 0, \forall i' \neq i. \quad (25)$$

Thus when $\pi_{ij} > \pi_{i'j}$, $q_{ij} > q_{i'j}$ holds.

From the above proof, it can be seen that content producers who contribute more will receive greater rewards. Therefore, content producers who receive the same rewards must have equivalent contributions, and vice versa. In other words, $\pi_{ij} = \pi_{i'j}$ if and only if $q_{ij} = q_{i'j}$. Therefore, the proof is finished. $\square$

We can summarize this property to **Proposition 2**.

Proposition 2: (Monotonicity of the Contributions): As a strictly increasing function of π , the contribution amount of content q satisfies:

$$0 \leq q_{1j} \leq \dots \leq q_{ij} \leq \dots \leq q_{Nj}. \quad (26)$$

Likewise, the third property of feasible contracts can be summarized in **Lemma 3**.

Lemma 3: $\forall j \in \mathcal{M}, \forall i, i' \in \mathcal{N}$ and $i \neq i'$, any feasible contract items satisfy $U^P(\theta_i, q_{ij}) > U^P(\theta_{i'}, q_{i'j})$ if and only if $\theta_i > \theta_{i'}$.

Proof:

1) *Sufficiency:* $\theta_i > \theta_{i'} \rightarrow U^P(\theta_i, q_{ij}) > U^P(\theta_{i'}, q_{i'j})$.

According to IC constraints, we have:

$$\begin{aligned} U^P(\theta_i, q_{ij}) &= \theta_i \ln(1 + \pi_{ij}) - cq_{ij} \\ &\geq \theta_i \ln(1 + \pi_{i'j}) - cq_{i'j} \\ &> \theta_{i'} \ln(1 + \pi_{i'j}) - cq_{i'j} = U^P(\theta_{i'}, q_{i'j}). \end{aligned} \quad (27)$$

Thus, $U^P(\theta_i, q_{ij}) > U^P(\theta_{i'}, q_{i'j})$ holds.

2) *Necessity:* $U^P(\theta_i, q_{ij}) > U^P(\theta_{i'}, q_{i'j}) \rightarrow \theta_i > \theta_{i'}$.

Due to $U^P(\theta_i, q_{ij}) > U^P(\theta_{i'}, q_{i'j})$ and IC constraints, we can easily obtain:

$$\begin{aligned} \theta_i \ln(1 + \pi_{ij}) - cq_{ij} &> \theta_{i'} \ln(1 + \pi_{i'j}) - cq_{i'j} \\ &\geq \theta_{i'} \ln(1 + \pi_{ij}) - cq_{i'j}. \end{aligned} \quad (28)$$

Therefore, $\theta_i > \theta_{i'}$ holds. The proof is completed. $\square$

From **Lemma 3**, it is proved that if $\theta_1 < \dots < \theta_i < \dots < \theta_N$, then $U^P(\theta_1, q_{1j}) < \dots < U^P(\theta_i, q_{ij}) < \dots < U^P(\theta_N, q_{Nj})$, which means a higher-type node can achieve greater utility compared to a lower-type node.

E. Optimal Contract Design Under Information Asymmetry

In this part, we employ the feasibility conditions of the contract obtained in IV-D to simplify the IR and IC constraints, respectively.

1) *IR Constraints Reduction:* Firstly, we simplify the IR constraints described in (13b), which require that the utility for each type of content producer must be greater than 0. Besides, according to **Lemma 3**, the content producer with the lowest type will obtain the minimum utility. Therefore, as long as the utility of θ_1 -type content producer satisfies the IR constraint, the other types of content producers must be able to meet the IR constraint [22]. Thus, (13b) can be simplified to:

$$\theta_1 \ln(1 + \pi_{1j}) - cq_{1j} \geq 0. \quad (29)$$

In order to maximize the utility of content consumer, the optimal contract (q_{1j}^*, π_{1j}^*) for θ_1 -type content producer must satisfy:

$$\theta_1 \ln(1 + \pi_{1j}^*) - cq_{1j}^* = 0. \quad (30)$$

Proof: We use proof by contradiction. Assuming $\theta_1 \ln(1 + \pi_{1j}^*) - cq_{1j}^* > 0$, then content consumer j , in order to increase its expected utility while guaranteeing the feasibility of the contract, can either choose to decrease π_{1j}^* or increase

q_{1j}^* until $\theta_1 \ln(1 + \hat{\pi}_{1j}) - c\hat{q}_{1j} = 0$ is satisfied. In this case, we have:

$$\theta_1 \ln(1 + \hat{\pi}_{1j}) - c\hat{q}_{1j} > \theta_1 \ln(1 + \pi_{1j}^*) - cq_{1j}^*, \quad (31)$$

which contradicts the assumption that (q_{1j}^*, π_{1j}^*) is the optimal contract. Therefore, optimal contract must satisfy $\theta_1 \ln(1 + \pi_{1j}^*) - cq_{1j}^* = 0$. The proof is finished. $\square$

2) *IC Constraints Reduction*: Then, the IC constraints given in (13c) are reduced. Firstly, four concepts similar to those in [21] and [36] are defined.

- downward incentive constraints (DICs): the IC between type- i content producer and type- i' , where $i' \in \{1, \dots, i-1\}$ and $i \neq i'$.
- local downward incentive constraints (LDICs): the IC between type- i content producer and type- $i-1$, where $i > 1$.
- upward incentive constraints (UICs): the IC between type- i content producer and type- i' , where $i' \in \{i+1, \dots, N\}$ and $i \neq i'$.
- local upward incentive constraints (LUICs): the IC between type- i content producer and type- $i+1$, where $i < N$.

Then, we prove that DICs can be substituted by LDICs.

Proof: For type- i , type- $i-1$ and type- $i-2$ content producer, there exists two LDICs as follows:

$$\theta_i \ln(1 + \pi_{ij}) - cq_{ij} \geq \theta_i \ln(1 + \pi_{(i-1)j}) - cq_{(i-1)j}, \quad (32)$$

$$\begin{aligned} \theta_{i-1} \ln(1 + \pi_{(i-1)j}) - cq_{(i-1)j} \\ \geq \theta_{i-1} \ln(1 + \pi_{(i-2)j}) - cq_{(i-2)j}. \end{aligned} \quad (33)$$

Due to $\theta_i < \theta_{i-1}$, from (32) and (33) we can obtain:

$$\begin{aligned} \theta_i [\ln(1 + \pi_{(i-1)j}) - \ln(1 + \pi_{(i-2)j})] \\ \geq \theta_{(i-1)} [\ln(1 + \pi_{(i-1)j}) - \ln(1 + \pi_{(i-2)j})] \\ \geq cq_{(i-1)j} - cq_{(i-2)j}. \end{aligned} \quad (34)$$

Therefore, (32) can be restated as:

$$\begin{aligned} \theta_i \ln(1 + \pi_{ij}) - cq_{ij} &\geq \theta_i \ln(1 + \pi_{(i-1)j}) - cq_{(i-1)j} \\ &\geq \theta_i \ln(1 + \pi_{(i-2)j}) - cq_{(i-2)j}, \end{aligned} \quad (35)$$

which can be summarized as

$$\begin{aligned} \theta_i \ln(1 + \pi_{ij}) - cq_{ij} &\geq \theta_i \ln(1 + \pi_{(i-1)j}) - cq_{(i-1)j} \\ &\geq \dots \geq \theta_i \ln(1 + \pi_{1j}) - cq_{1j}. \end{aligned} \quad (36)$$

Thus, DICs can be reduced to LDICs. The proof is finished. $\square$

Similarly, UICs can be substituted by LUICs:

$$\begin{aligned} \theta_i \ln(1 + \pi_{ij}) - cq_{ij} &\geq \theta_i \ln(1 + \pi_{(i+1)j}) - cq_{(i+1)j} \\ &\geq \dots \geq \theta_i \ln(1 + \pi_{Nj}) - cq_{Nj}. \end{aligned} \quad (37)$$

Due to space constraints, the proof process is omitted.

To sum up, the IC constraints can be reduced as:

$$\begin{aligned} \theta_i \ln(1 + \pi_{ij}) - cq_{ij} &\geq \theta_i \ln(1 + \pi_{(i-1)j}) - cq_{(i-1)j}, \\ i &= 2, \dots, N, j \in \mathcal{M}. \end{aligned} \quad (38)$$

After transforming, we have:

$$q_{ij} \leq q_{(i-1)j} + \frac{\theta_i}{c} \ln\left(\frac{1 + \pi_{ij}}{1 + \pi_{(i-1)j}}\right), \quad i = 2, \dots, N, j \in \mathcal{M}. \quad (39)$$

In order to increase the expected utility and maintain the feasibility of the contract, content consumer j must either decrease π_{ij} or increase q_{ij} until the equality holds. Therefore, for type- θ_i content producer, the optimal contract (q_{ij}^*, π_{ij}^*) should meet:

$$q_{ij}^* = q_{(i-1)j}^* + \frac{\theta_i}{c} \ln\left(\frac{1 + \pi_{ij}^*}{1 + \pi_{(i-1)j}^*}\right), \quad i = 2, \dots, N, j \in \mathcal{M}. \quad (40)$$

Therefore, based on the above constraint simplification, our initial optimization problem **P1** can be reformulated as:

$$\mathbf{P2}: \max_{q_{ij}, \pi_{ij}} \sum_{i=1}^N p_i(\sqrt{aq_{ij}} - \pi_{ij}), \quad (41)$$

$$\text{s.t. } q_j^{\min} \leq q_{ij} \leq q_j^{\max}, \quad (41a)$$

$$IR: \theta_1 \ln(1 + \pi_{1j}) - cq_{1j} = 0, \quad (41b)$$

$$\begin{aligned} IC: \theta_i \ln(1 + \pi_{ij}) - cq_{ij} &= \theta_i \ln(1 + \pi_{(i-1)j}) \\ &- cq_{(i-1)j}, \end{aligned} \quad (41c)$$

$$0 \leq \pi_{1j} \leq \dots \leq \pi_{ij} \leq \dots \leq \pi_{Nj}. \quad (41d)$$

Further simplify (41b) and (41c), we can obtain:

$$q_{1j} = \frac{\theta_1 \ln(1 + \pi_{1j})}{c}, \quad (42)$$

$$q_{ij} = q_{1j} + \sum_{i=2}^N \frac{\theta_i}{c} \ln\left(\frac{1 + \pi_{ij}}{1 + \pi_{(i-1)j}}\right), \quad \forall i = 2, \dots, N. \quad (43)$$

We can transform **P2** into an equivalent problem:

$$\mathbf{P3}: \min_{q_{ij}, \pi_{ij}} \sum_{i=1}^N -p_i(\sqrt{aq_{ij}} - \pi_{ij}), \quad (44)$$

$$\text{s.t. } q_j^{\min} \leq q_{ij} \leq q_j^{\max}, \quad (44a)$$

$$IR: cq_{1j} - \theta_1 \ln(1 + \pi_{1j}) = 0, \quad (44b)$$

$$\begin{aligned} IC: cq_{ij} - \theta_i \ln(1 + \pi_{ij}) &= cq_{(i-1)j} \\ &- \theta_i \ln(1 + \pi_{(i-1)j}), \end{aligned} \quad (44c)$$

$$0 \leq \pi_{1j} \leq \dots \leq \pi_{ij} \leq \dots \leq \pi_{Nj}. \quad (44d)$$

We set the optimization problem (44) as F , and can easily obtain the Hessian matrix of the optimization problem as follows:

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 F}{\partial q^2} & \frac{\partial^2 F}{\partial q \partial \pi} \\ \frac{\partial^2 F}{\partial \pi \partial q} & \frac{\partial^2 F}{\partial \pi^2} \end{bmatrix} = \begin{bmatrix} \frac{\partial^2 F}{\partial q^2} & 0 \\ 0 & \frac{\partial^2 F}{\partial \pi^2} \end{bmatrix} = \begin{bmatrix} \frac{1}{4} paq^{-\frac{3}{2}} & 0 \\ 0 & 0 \end{bmatrix}. \quad (45)$$

Given that the eigenvalues of the Hessian matrix are all greater than or equal to 0, it follows that the objective function

is convex. Furthermore, since (44a) and (44d) are linear constraints, they form convex sets. Similarly, it is easy to prove that (44b) and (44c) are convex constraints, therefore the problem is a convex optimization problem. Thus, we can solve it by the Lagrange multiplier method. The corresponding Lagrangian function $\mathcal{L}$ can be written as:

$$\begin{aligned} \mathcal{L}(\{q_{ij}\}, \{\pi_{ij}\}, \{\alpha_i\}, \{\beta_i\}, \{\gamma_i\}, \{\eta_i\}) \\ = \sum_{i=1}^N p_i(\sqrt{aq_{ij}} - \pi_{ij}) + \sum_{i=1}^N \alpha_i(q_{ij} - q_j^{\min}) \\ - \sum_{i=1}^N \beta_i(q_{ij} - q^{\max}) + \gamma_1[\theta_1 \ln(1 + \pi_{1j}) - cq_{1j}] \\ + \sum_{i=2}^N \gamma_i \left[\frac{\theta_i}{c} (\ln(1 + \pi_{ij}) - \ln(1 + \pi_{(i-1)j})) + q_{(i-1)j} - q_{ij} \right] \\ + \eta_1 \pi_{1j} + \sum_{i=2}^N \eta_i (\pi_{ij} - \pi_{(i-1)j}), \end{aligned} \quad (46)$$

where $\{\alpha_i, i = 1, \dots, N\}$, $\{\beta_i, i = 1, \dots, N\}$, $\{\gamma_i, i = 1, \dots, N\}$, $\{\eta_i, i = 1, \dots, N\}$ are Lagrange multipliers. The KKT conditions are demonstrated as follows:

- Primal constraints:

$$\begin{cases} q_{ij} \geq q_j^{\min}, \forall i \in \mathcal{N}, \forall j \in \mathcal{M}; \\ q_{ij} \leq q^{\max}, \forall i \in \mathcal{N}, \forall j \in \mathcal{M}; \\ \pi_{1j} \geq 0, \forall j \in \mathcal{M}; \\ \pi_{ij} \geq \pi_{(i-1)j}, \forall i \in \mathcal{N}, i \neq 1, \forall j \in \mathcal{M}; \\ \theta_1 \ln(1 + \pi_{1j}) - cq_{1j} = 0, \forall j \in \mathcal{M}; \\ \theta_i \ln(1 + \pi_{ij}) - cq_{ij} = \theta_i \ln(1 + \pi_{(i-1)j}) - cq_{(i-1)j}, \\ \quad \forall i \in \mathcal{N}, i \neq 1, \forall j \in \mathcal{M}. \end{cases} \quad (47)$$

- Dual constraints:

$$\alpha_i \geq 0, \beta_i \geq 0, \gamma_i \geq 0, \eta_i \geq 0, \forall i \in \mathcal{N}, \forall j \in \mathcal{M}. \quad (48)$$

- Complementary slackness:

$$\begin{cases} \alpha_i(q_{ij} - q_j^{\min}) = 0, \forall i \in \mathcal{N}, \forall j \in \mathcal{M}; \\ \beta_i(q_{ij} - q^{\max}) = 0, \forall i \in \mathcal{N}, \forall j \in \mathcal{M}; \\ \eta_1 \pi_{1j} = 0, \forall j \in \mathcal{M}; \\ \eta_i (\pi_{ij} - \pi_{(i-1)j}) = 0, \forall i \in \mathcal{N}, i \neq 1, \forall j \in \mathcal{M}; \\ \gamma_1[\theta_1 \ln(1 + \pi_{1j}) - cq_{1j}] = 0, \forall j \in \mathcal{M}; \\ \gamma_i \left(\frac{\theta_i}{c} \cdot \ln \frac{1 + \pi_{ij}}{1 + \pi_{(i-1)j}} + q_{(i-1)j} - q_{ij} \right) = 0, \\ \quad \forall i \in \mathcal{N}, i \neq 1, \forall j \in \mathcal{M}. \end{cases} \quad (49)$$

- The first-order conditions of the Lagrangian:

$$\begin{cases} \frac{\partial L}{\partial q_{ij}} = \frac{p_i}{2\sqrt{aq_{ij}}} + \alpha_i - \beta_i - \gamma_i + \gamma_{i+1} = 0, \\ \quad \forall i \in \mathcal{N}, i \neq N, \forall j \in \mathcal{M}; \\ \frac{\partial L}{\partial q_{Nj}} = \frac{p_N}{2\sqrt{aq_{Nj}}} + \alpha_N - \beta_N - \gamma_N = 0, \\ \quad \forall j \in \mathcal{M}; \\ \frac{\partial L}{\partial \pi_{ij}} = -p_i + \frac{1}{1 + \pi_{ij}} \left(\frac{\gamma_i \theta_i}{c} - \frac{\gamma_{i+1} \theta_{i+1}}{c} \right) \\ + \eta_i - \eta_{i+1} = 0, \forall i \in \mathcal{N}, i \neq N, \forall j \in \mathcal{M}; \\ \frac{\partial L}{\partial \pi_{Nj}} = -p_N + \frac{1}{1 + \pi_{Nj}} \cdot \frac{\gamma_N \theta_N}{c} + \eta_N = 0, \\ \quad \forall j \in \mathcal{M}. \end{cases} \quad (50)$$

According to the KKT conditions above, optimal contract under information asymmetry must satisfy:

$$q_{ij} = \begin{cases} \frac{p_i^2}{4a(\gamma_i - \gamma_{i+1} + \beta_i - \alpha_i)^2}, & i \neq N, \\ \frac{p_N^2}{4a(\gamma_N + \beta_N - \alpha_N)^2}, & i = N. \end{cases} \quad (51)$$

$$\pi_{ij} = \begin{cases} \frac{\gamma_i \theta_i - \gamma_{i+1} \theta_{i+1}}{c(p_i + \eta_{i+1} - \eta_i)} - 1, & i \neq N, \\ \frac{\gamma_N \theta_N}{c(p_N - \eta_N)} - 1, & i = N. \end{cases} \quad (52)$$

F. Optimal Contract Design Without Information Asymmetry

In the scenario with complete information, we assume that there exists a selfish content consumer who greatly knows the type of each content producer, and will increase its profit by utilizing the complete information. In this case, content producers are unable to misrepresent their types, thus only the IR constraints need to be satisfied.

Specifically, the content consumer will decrease π_{ij} while making the contract meet IR constraints until the equality holds. Therefore, the optimal contract (q_{ij}^*, π_{ij}^*) with complete information must satisfy $\theta_i \ln(1 + \pi_{ij}^*) - cq_{ij}^* = 0, \forall i \in \mathcal{N}, \forall j \in \mathcal{M}$.

The optimization problem without information asymmetry can be written as

$$\mathbf{P4}: \max_{q_{ij}, \pi_{ij}} \sum_{i=1}^N p_i(\sqrt{aq_{ij}} - \pi_{ij}), \quad (53)$$

$$\text{s.t. } q_j^{\min} \leq q_{ij} \leq q^{\max}, \quad (53a)$$

$$IR: \theta_i \ln(1 + \pi_{ij}) - cq_{ij} = 0, \quad (53b)$$

$$0 \leq \pi_{1j} \leq \dots \leq \pi_{ij} \leq \dots \leq \pi_{Nj}. \quad (53c)$$

which can also be addressed using the method of Lagrange multipliers.

V. CONSUMER-PRODUCER MATCHING ALGORITHM

The optimal contracts derived from the previous section provide rewards and data volumes for each consumer-producer pair, and the types of content producers are also exposed to content consumers. However, in practice, there may be

multiple content consumers requesting the same content at the same time. The optimal content producer for each content consumer is the one with the highest type, also known as type N , whom they are most eager to sign contracts with. Thus, it is necessary to make a reasonable and precise match between UAVs on both sides.

A. Matching Definition

In this subsection, we first provide an introduction to some key components of the matching game, and subsequently, we present the definition of the many-to-one matching game in UNDN.

We define the consumer-producer matching game by the following components:

- The set of content consumers $\mathcal{M}$;
- The set of content producers $\mathcal{N}$;
- The preference lists $\mathcal{P}^C, \mathcal{P}^P$ for content consumers and content producers, which is generated by sorting the utility values.

Definition 5: We define the matching function Φ as a mapping from $\mathcal{M} \cup \mathcal{N} \rightarrow \mathcal{M} \cup \mathcal{N}$, where each content consumer $j \in \mathcal{M}$ and content producer $i \in \mathcal{N}$ satisfy the following conditions:

- 1) $\Phi(j) \subset \mathcal{N}$ and $|\Phi(j)| \leq 1$;
- 2) $\Phi(i) \subset \mathcal{M}$ and $|\Phi(i)| \leq K_i$;
- 3) $\Phi(j) = i$ if and only if $\Phi(i) = j$,

where K_i denotes the quota of content producer i and is determined by the residual battery B_i :

$$K_i = B_i K_{\max}. \quad (54)$$

$K_{\max}$ is the maximum number of matches in the scene.

Conditions 1) and 2) imply that each content consumer can match with at most one content producer, while each content producer can match with multiple content consumers. This defines a many-to-one matching scenario. Condition 3) illustrates that if content consumer j is matched with content producer i , then i is also matched with j .

B. Preference List Construction

Given that real content transmission occurs after matching, it is crucial for content producers to factor in the transmission cost and update the utility functions of both parties throughout the matching process.

For content consumer, after completing the contract design phase, the type of content producer and the selected contract are fully exposed. Each content consumer hopes to choose a content producer with a higher type, as such a content producer can provide more content and has higher reputation value and residual battery, which enables better completion of transmission tasks. Therefore, this type of content producer deserves more rewards. The utility function of content consumers in the matching stage are given as follows:

$$U_m^C(\theta_i, q_{ij}) = \varepsilon \ln(1 + q_{ij}) - \delta_i, \quad (55)$$

For a fixed type of content producer, the rewards they can receive from different content consumers are the same.

Algorithm 1 Consumer-Producer Matching Algorithm

Input: Utility matrix of Content Consumer U_m^C , utility matrix of content producer U_m^P , matching capacity matrix K

Output: Matching results Φ

```

1: Initialization:
2: Construct the preference lists of content consumers  $P^C$ 
3: Construct the preference lists of content producers  $P^P$ 
4: Content consumers propose to content producers:
5: for all content consumers who are not matched do
6:   Propose to the content producer at the top of preference list
7: end for
8: Content producers make decisions:
9: for all content producers  $i \in \mathcal{N}$  do
10:  if  $|\Phi(i)| \leq K_i$  then
11:    content producer  $i$  accepts all the proposed content consumers
12:  else
13:    content producer  $i$  accepts the most preferred  $K_i$  content consumers and rejects the rest ones
14:  end if
15: end for
16: if there is still a consumer has not been matched then
17:   Go to step 4
18: else
19:   Go to step 21
20: end if
21: End of algorithm

```

Therefore, it is necessary to consider the issue of their own transmission costs. They tend to choose content consumers who can reduce their transmission costs to save energy. Thus, the utility function of content producers in the matching stage are given as follows:

$$U_m^P(\theta_i, q_{ij}) = \delta_i - \lambda E_{ij}^T, \quad (56)$$

where $\delta_i = m\theta_i$ represents the reward provided by content consumer j to θ_i -type content producer. λ denotes the unit transmission energy cost, and ε is a positive weighting factor.

Based on the outcomes of the contract, content consumers and content producers calculate the utilities according to (55) and (56), then establish their preference lists $\mathcal{P}^C, \mathcal{P}^P$ by sorting utility values in descending order. Specifically, we define a symbol $\succeq$ to express the preference relation between UAVs. For example, $i \succeq_j i'$ means that j prefers i to i' or considers them equally preferable.

The content producer seeks increased rewards and reduced transmission energy consumption to maximize utility. Hence, it favors content consumers who can contribute to the higher utility. Therefore, for $\forall i \in \mathcal{N}, \forall j, j' \in \mathcal{M}$, we have:

$$j \succeq_i j' \iff U_m^P(\theta_i, q_{ij}) > U_m^P(\theta_i, q_{ij'}). \quad (57)$$

Similarly, for a content consumer, it intends to obtain more content and provide less reward to achieve greater utility, so it prefers the content producer who can make it achieve greater

TABLE II
PARAMETERS

Parameter	Value	Parameter	Value
Number of content producers N	10, 20 ~ 100	Energy consumption factor of free space model η_{fs}	$10pJ/(bit * m^2)$ [38]
Number of content consumers M	1, 5, 15, 20 ~ 100	Energy consumption factor of multi-path fading model η_{mp}	$0.0013pJ/(bit*m^4)$ [38]
Probability of type- θ_i producer p_i	$\frac{1}{N}$	Maximum number of matching K_{max}	5
Weight factors a, ε, m	50, 1, 2	Content size q	$[0, 20]Mb$
Caching cost per Mb c	0.1	Reputation value R	$[0, 1]$
Transmission cost per bit λ	0.5	Residual battery B	$[0, 1]$
Energy consumption per bit E_{elec}	$50nJ/bit$ [38]	Weight factors α_1, α_2	0.6, 0.4

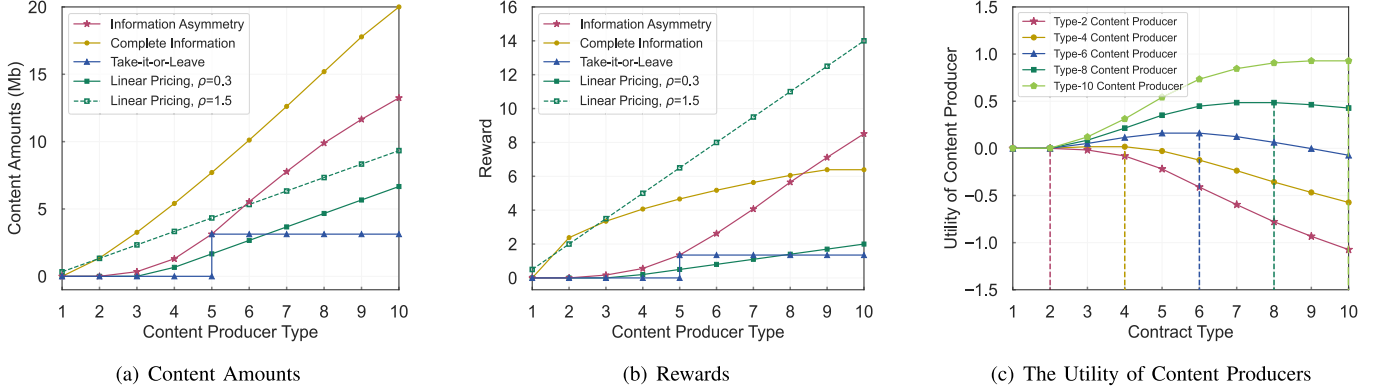


Fig. 3. Contract feasibility.

utility. Therefore, for $\forall i, i' \in \mathcal{N}, \forall j \in \mathcal{M}$, we have:

$$i \succeq_j i' \iff U_m^C(\theta_i, q_{ij}) > U_m^C(\theta_{i'}, q_{i'j}). \quad (58)$$

C. GS Matching Algorithm

In this subsection, we will provide specific steps for the GS matching algorithm.

- 1) **Construct preference lists:** Content consumers and content producers calculate their utilities separately and then rank the utility values in descending order to establish their own preference lists.
- 2) **Both sides make choices:** Each content consumer proposes a matching request to the top nodes in its preference list, while each content producer accepts the top K items and rejects the remaining content consumers.
- 3) **Both sides reselect:** The rejected content consumer removes the rejecting content producer from its preference list and repeats step 2) until either all content consumers are matched with content producers or all content producers reach their maximum capacity for matching.

The matching process is summarized in Algorithm 1.

D. Stability and Pareto Optimality Analysis

In this subsection, we will give the stability and Pareto optimality analysis of the consumer-producer matching algorithm. A stable matching means that none of the participants have motivations to change their partners. The definition of stability is shown as follows.

Definition 6 (Stability): A matching Φ is stable if there exists no blocking pairs [26], [37]. A matching is blocked by a pair (i, j) if:

- 1) blocking occurs simultaneously in i and j ;
- 2) i or j chooses each other will result in a higher utility compared to the currently matched partners, i.e.,

$$\begin{aligned} \exists i \in \mathcal{N}, j \in \mathcal{M}, \Phi(j) = i^*, \Phi(i) = j^*, i^* \neq i, j^* \neq j, \\ U_m^P(\theta_i, q_{ij}) > U_m^P(\theta_i, q_{ij^*}) \\ \text{and } U_m^C(\theta_i, q_{ij}) > U_m^C(\theta_{i^*}, q_{i^*j}) \end{aligned} \quad (59)$$

Condition 1) illustrates that pair blocking occurs in pairs, not in a single party, and condition 2) explains that the blocking pair must have a higher utility than the current matching pair.

Given Definition 6 and Algorithm 1, we present **Lemma 4** below and prove that our matching algorithm is stable.

Lemma 4: Our proposed consumer-producer matching algorithm is a stable matching.

Proof: As we discussed before, a stable matching is essentially characterized by the absence of blocking pairs. Therefore, we verify that there are no blocking pairs in the matching algorithm. The preference lists of content consumers and content producers are constructed according to (57) and (58), thus are strictly monotone increasing. Furthermore, we can observe from line 6 and lines 10~13 in Algorithm 1, the content consumer proposes to the preferred content producer, and the content producer accepts the top K content consumers among the proposed content consumers based on its

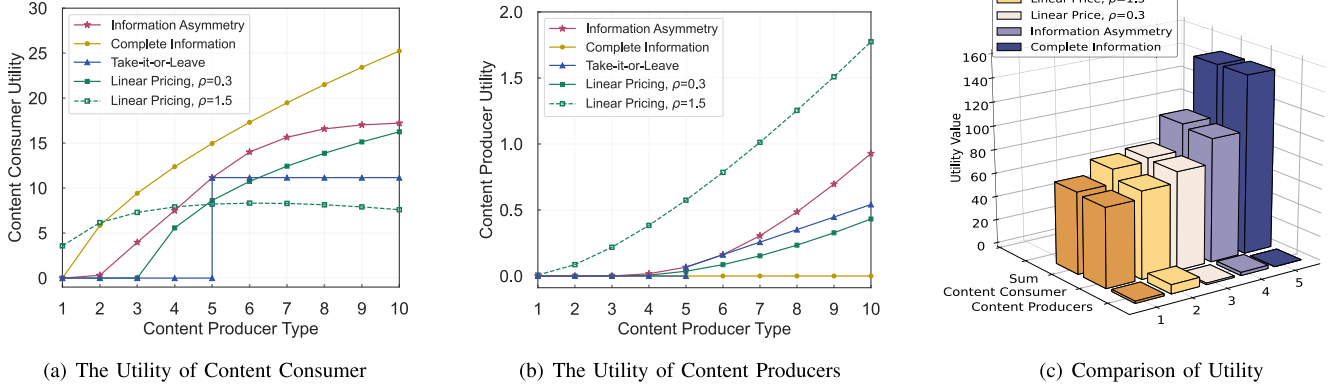


Fig. 4. Contract efficiency.

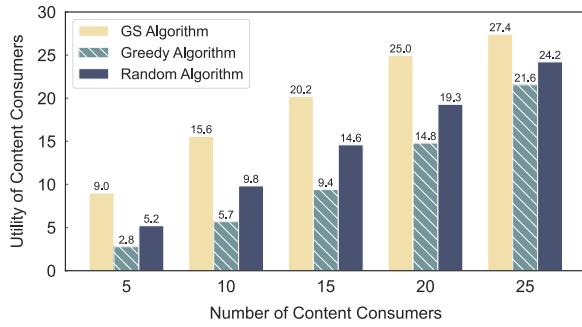


Fig. 5. Utility of content consumers.

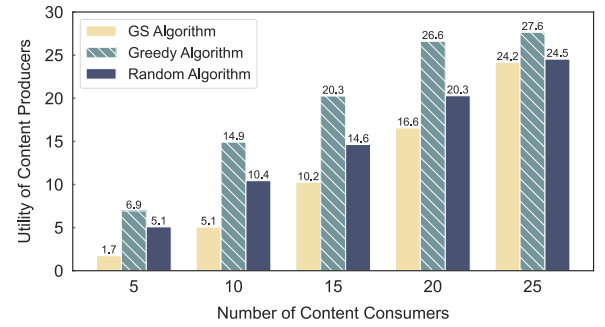


Fig. 6. Utility of content producers.

preference list. This ensures the absence of any blocking pairs, and therefore the matching is stable. The proof is finished. $\square$

Based on the properties of stable matching, we can obtain **Lemma 5** as follows.

Lemma 5: *The obtained matching result set Φ is weak Pareto optimal for each content consumer $j \in \mathcal{M}$ and content producer $i \in \mathcal{N}$.*

Proof: This proof process largely follows [28] and [30]. In our proposed consumer-producer matching algorithm, each participant selects matching objects based on the ranking of the preference list, which indicates that the current choices of both parties are the positions available in the preference lists with the highest priority. That is to say, they cannot achieve higher utility by selecting other matching objects. This is consistent with the matching stability we demonstrated in **Lemma 4**. Therefore, there is no Pareto improvement throughout the matching process, indicating that the matching algorithm is weak Pareto optimal. The proof is finished. $\square$

E. Complexity Analysis

This section provides a complexity analysis of the proposed mechanism. In the first stage, i.e., the contract-based incentive mechanism, solving the optimal contract (44) is a convex optimization problem. Since there are N contract items (q, π) , the number of equality constraints is N and the number of inequality constraints is $3N$. Therefore, the computation complexity is $\mathcal{O}(N)$. As we discussed before, there are M

content consumers in the system, and all of them need to design contracts, thus the total computation complexity is $\mathcal{O}(MN)$.

In the second stage, i.e., the matching mechanism, the matching results can be derived after the GS algorithm. For a content consumer, the complexity of constructing a preference list of N content producers with an efficient sorting algorithm is $\mathcal{O}(N \log N)$ [21]. Therefore, the computing complexity can be $\mathcal{O}(MN \log N)$ for M content consumers. Similarly, N content producers can build the preference lists with a computing complexity of $\mathcal{O}(NM \log M)$. Then, the input size of GS algorithm is $\sum_{i \in \mathcal{N}} |P_i^C| + \sum_{j \in \mathcal{M}} |P_j^P| = 2MN$, where $|P|$ represents the length of the preference list, and the computing complexity of the loop in Algorithm 1 is $\mathcal{O}(MN)$. Therefore, the overall computing complexity of the algorithm is $\mathcal{O}(MN(\log MN + 1)) = \mathcal{O}(MN \log MN)$.

VI. SIMULATION RESULTS AND ANALYSIS

In this section, we evaluate the proposed mechanism through multiple sets of simulation experiments. We use *Python* to address the convex optimization problem. The parameter values used for simulation can be found in Table II.

A. Contract Feasibility

We consider one content consumer requests a kind of content while 10 content producers can offer the content. As mentioned before, the number of content producers is

assumed to be equal to the number of contract types, and the type distribution of content producers follows a uniform distribution, i.e., $p_i = \frac{1}{N}, \forall i \in \mathcal{N}$. We conduct experiments under the situations with and without information asymmetry and compare them with two other algorithms: *linear pricing* [36] and *take-it-or-leave contract* [39]. *Linear pricing* is a mechanism employed under information asymmetry. In this method, the content consumer sets the reward proportional to content amounts, which obeys $\pi = \rho q$ and ρ is the unit reward. In this paper, inspired by [21], [40], and [41], we set ρ as 0.3 and 1.5. In *take-it-or-leave contract*, there exists a type threshold θ_{th} . Content consumer designs a set of same contract items (q_{th}, π_{th}) based on θ_{th} . And a content producer with type greater than this threshold, i.e., $\forall \theta_i \in \Theta, \theta_i \geq \theta_{th}$ will accept and sign the contracts, while a content producer with type lower than this threshold, i.e., $\forall \theta_i \in \Theta, \theta_i < \theta_{th}$ will reject and discard the contracts. In this paper, θ_{th} is set as θ_5 followed by [22]. Fig. 3 shows the verification results of contract feasibility. The shared content amounts and corresponding rewards of the four algorithms are shown in Fig. 3(a) and Fig. 3(b), respectively. As we can see from these two figures, the content amounts and the corresponding rewards all increase with the type of content producer, which has been proven in **Lemma 1** and **Lemma 2**. Moreover, the numerical results indicate that in the situation of complete information, content producers provide the most content and get the most reward, while the case of information asymmetry shows the middle content contributions and reward. In the case of *linear pricing*, the contributed content amounts and rewards both increase with ρ . And when $\rho = 1.5$, content producers with a type less than 5 can provide a greater amount of content compared to situations with information asymmetry. However, when the type is greater than 5, the situation reverses. This is because that content producers are regarded as the same type in *linear pricing*, the content-providing ability of high-type content producers has not been well utilized. In the case of *take-it-or-leave contract*, as we discussed before, content producers with type less than θ_5 will refuse the contracts, thereby not providing content and receiving no rewards, while content producers with types greater than or equal to θ_5 will accept the same contract items, thus providing the same content amounts and receiving the same rewards. It is worth noting that in Fig. 3(b), content producers with type greater than or equal to θ_8 contribute more content but receive fewer rewards in the case of complete information, compared to the information asymmetry situation. This is because in the case of complete information, the utilities of content producers are always 0, while the utilities are increased with types in the case of information asymmetry.

Fig. 3(c) shows the utilities of θ_2 -type, θ_4 -type, θ_6 -type, θ_8 -type and θ_{10} -type content producers when they choose different contract items. As we can see, $U^P(\theta_{10}) > U^P(\theta_8) > U^P(\theta_6) > U^P(\theta_4) > U^P(\theta_2)$, which means the higher type can obtain higher utilities and is consistent with **Lemma 3**. Furthermore, Fig. 3(c) demonstrates that each type of content producer can achieve maximum utility only when selecting the contract item designed for its type, which conforms to the IC constraints of the optimal contracts.

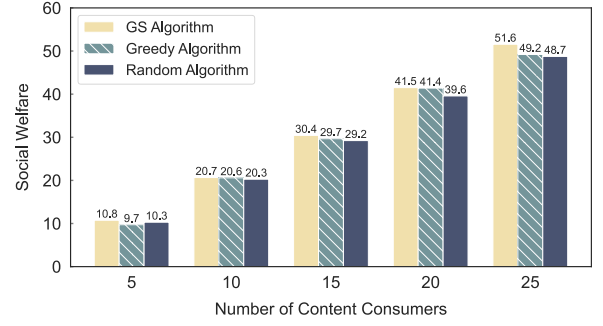


Fig. 7. Social welfare.

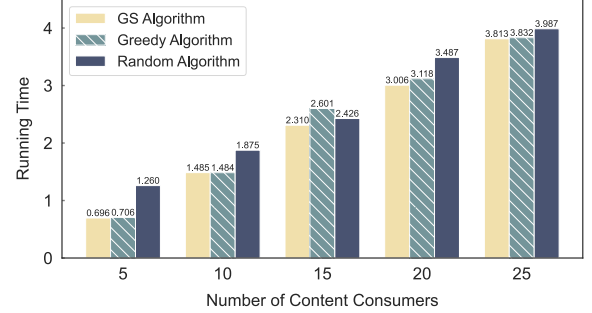
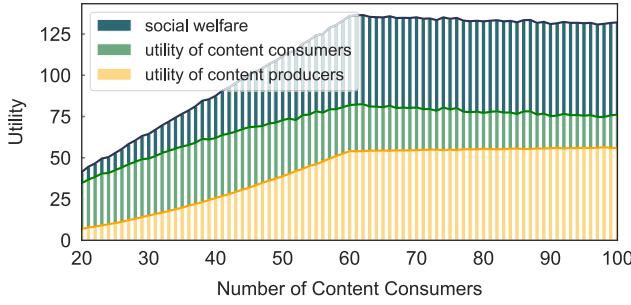


Fig. 8. Running time.

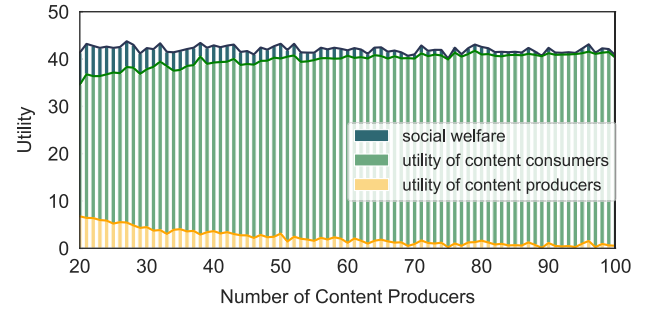
B. Contract Efficiency

Next, the verification results of contract efficiency are presented in Fig. 4. Fig. 4(a) shows how the utility of the content consumer varies with types. We can observe that the content consumer can achieve higher utility by choosing content producers with larger types. In addition, the content consumer with complete information can achieve the highest utility values, since it can perfectly utilize the information to maximize its own benefits. Besides, In the case of information asymmetry, the utility of content consumer is lower compared to that in the scenario with complete information. This is due to the fact that asymmetric information, to some extent, protects content producers from being excessively exploited. Meanwhile, in the *linear pricing* mechanism, when $\rho = 0.3$, the utility of content consumer is lower than the situation of asymmetric information due to the lowest content amounts. While when $\rho = 1.5$, the utility of content consumer is higher than the situation of asymmetric information when the chosen type of content producer θ is less than 4. This is because more content amounts dominates when $\theta \leq 4$, while more reward dominates when $\theta > 4$. Furthermore, according to the principles of *take-it-or-leave contract*, content producers with type less than θ_5 refuse the contracts and therefore do not offer content to the content consumer, resulting in a utility of 0. Conversely, content producers with types greater than or equal to θ_5 accept the same contract items, thus making the utility of content consumers unchanged.

Fig. 4(b) presents the utility of content producers changes with types. We can observe that in the *linear pricing* algorithm when $\rho = 1.5$, content producers can obtain the highest utility due to the highest rewards. In the case of information



(a) 20 content producers with the number of content consumers changes
Fig. 9. Changes in the utility of content consumers and content producers.



(b) 20 content consumers with the number of content producers changes

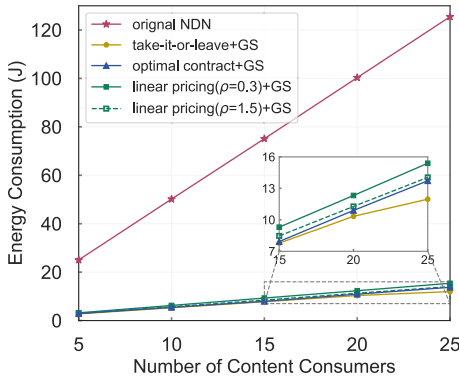


Fig. 10. Energy consumption.

asymmetry, content producers can obtain the second highest utility due to the protection of incomplete information. As we elaborated earlier, the contract only needs to obey IR constraints, and the utilities of content producers meet no less than zero under complete information. Therefore, the utilities of different types of content producers are 0. As for the *linear pricing* mechanism when $\rho = 0.3$, the utility of content producers lies in the middle. Additionally, in *take-it-or-leave contract*, only content producers with type greater than θ_5 will accept the contract, and thus these content producers can achieve non-zero utilities.

Fig. 4(c) demonstrates the utility of the content consumer, the entire utility of content producers, and the social welfare of these four algorithms. Simulation results show that contracts under information asymmetry can achieve higher social welfare while balancing the utility of both sides.

C. Matching Algorithm Evaluation

In this part, We first compare the Gale-Shapley matching algorithm with the other two algorithms: *greedy algorithm* and *random algorithm*. In *greedy algorithm*, the content producer with a higher type has priority to choose content consumers to serve until the quota reaches its maximum K . Next, it is the turn of content producers with lower types to choose. In *random algorithm*, content consumers and content producers are randomly matched to each other. The performance of *random algorithm* is assessed through averaging over 500 iterations. We consider that at a certain

time slot, multiple content consumers simultaneously request the same content while 10 content producers can provide such content. It's important to note that the minimum amount of content q^{min} each content consumer can accept varies. The matching phase is triggered after the contract phase under asymmetry information is completed.

Fig. 5 shows the utility of content consumers of GS algorithm, *greedy algorithm* and *random algorithm*, from which we can observe that the utilities increase with the number of content consumers. Besides, the case of the GS algorithm obtains the most utility values among the three, while the case of the *random algorithm* gets the least. This is because, in the GS algorithm, content consumers propose to their preferred content producers based on utility functions, and the matching process is led by content consumers. Therefore, it is more possible to arrive at results that are more beneficial to content consumers.

Fig. 6 presents the utility of content producers of the three algorithms mentioned above. Firstly, we can observe that as the number of content consumers increases, the utility of content producers increases. Besides, it is evident that the *greedy algorithm* can reach the most utility of content producers, and the utility in GS algorithm is higher than in *random algorithm*. As we explained earlier, in the *greedy algorithm*, the high-type content producer is fully loaded before moving to the next one. Each content producer selects the top few content consumers in the preference list constructed based on its utility function, which means the matching process is led by content producers. Therefore, the *greedy algorithm* can yield results that are profitable for content producers.

Then, we compare the social welfare of these three algorithms in Fig. 7. The social welfare of the matching process increases with the number of content consumers. Furthermore, it is demonstrated that the GS algorithm has the best performance among the three.

The running time of different algorithms is also simulated as depicted in Fig. 8. The experimental results indicate that with an increase in the number of content consumers, the running time becomes longer, and overall, the GS algorithm has the shortest time consumption. In summary, the GS algorithm can achieve outstanding performance among the three algorithms.

Finally, we increase the network size and evaluate the performance of the GS algorithm in Fig. 9. Two scenarios are considered: the number of content producers is fixed

to 20 while the number of content consumers increases from 20 to 100, as shown in Fig. 9(a), and the number of content consumers is fixed to 20, the number of content producers increases from 20 to 100, as shown in Fig. 9(b). In Fig. 9(a), we can observe that as the number of content consumers grows, the utility of both sides and the social welfare also increase until the number of content consumers reaches around 60. This is because the maximum matching number of all content producers is around 60. When the number of content consumers exceeds this limit, they start competing with each other for content, which causes their utilities to decline. In Fig. 9(b), we can obtain that as the number of content producers increases, their utility gradually decreases, but the social welfare remains basically stable. This is because some content producers can already meet the content needs of all content consumers, and an increase in the number of content producers intensifies competition among them, leading to a decrease in their utility.

D. The Energy Efficiency of the Proposed Mechanism

In this part, we evaluate the energy efficiency of our proposed mechanism by comparing it with other algorithms and the original content-sharing mechanism of NDN. We fix the number of content producers at 10, vary the number of content consumers, and calculate the energy consumption of content producers when all content consumers request content once. As shown in Fig. 10, we observe that after conducting many-to-one matching, the transmission energy consumed by content producers is significantly reduced due to the absence of redundant data packet transmission. From the comparative experiments, it can be seen that our proposed mechanism demonstrates good energy efficiency performance. Additionally, the *take-it-or-leave* algorithm with GS matching shows the lowest energy consumption due to its low contributed content amounts and the content producers who signed the contracts being unable to meet the content provision needs of all content consumers.

VII. CONCLUSION

In this paper, we proposed a novel resource-efficient content sharing approach in UNDN combining contract theory and matching theory. We utilized the NDN architecture in UAV swarm networks to address the connection problem in the IP-based solutions. For the first time, we mainly addressed the issue of content redundancy during the content backhaul process in UNDN. Specifically, we reduce redundant content packets in the network and consequently decrease energy consumption by conducting many-to-one matching between content producers and content consumers. To establish preference lists for both sides, content consumers design optimal sets of contracts to identify the types of content producers and incentivize them to share content. Finally, extensive experimental results exhibit the superiority of our proposed algorithm concerning shared content amount, social welfare, runtime, energy consumption, and other aspects.

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